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BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

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ODING CLUB BMSIT × ALTERINO CLUB BMSIT

NRMAAN2026

Build. Innovate. Impact

Team Name: RehabGrip

Team Leader: Khush Chadha

Chosen Track: HEALTHTECH & BIO-WEARABLES

Problem Statement: Home hand rehabilitation lacks real-time supervision and objective progress tracking between physiotherapy visits

Powered By:

MASTRY HUB × Reskill

Problem Statement & Real-World Impact

BRIDGING THE GAP BETWEEN THERAPY SESSIONS FOR A STRONGER TOMORROW

Small Movements **Big Possibilities**

1 THE PROBLEM



Home hand exercises prescribed between physiotherapy visits (spaced 1–2 weeks apart)

Zero objective tracking during this gap — no rep count, no movement quality data

Progress relies on subjective patient self-report — unreliable, infrequent

2 TARGET AUDIENCE

Primary:
Stroke / hand-injury / post-surgical patients doing home exercises

Secondary:
Physiotherapists & occupational therapists prescribing the program

Institutional:
Outpatient clinics, home-care & tele-rehab, community health programs

3 CURRENT GAPS & IMPACT

40–70%
of stroke survivors globally have access to rehab services

63%+
of people needing rehab care do not receive it

No visibility into adherence, rep count, or movement quality between visits

Delays therapy adjustments — undermines the repetitive practice recovery depends on

Proposed Solution & Innovation

SIMPLE TECHNOLOGY. MEANINGFUL RECOVERY.

Small Movements **Big Possibilities**

1 CORE SOLUTION



- Wearable glove combining flex sensors (finger movement) + surface EMG (muscle activation)
- ESP32 acquires both signals in real time during a prescribed hand exercise
- Lightweight classifier (Random Forest/SVM) flags each rep as correct or deviating
- Instant haptic/visual feedback + logged session data for the therapist to review

2 INNOVATION FACTOR

Same Capabilities.
Radically More Accessible.

RehabGrip		Commercial Sensing Gloves
₹6,500	vs	₹1L+
RehabGrip's BOM is ~₹5,500–6,500		Commercial sensing gloves cost ₹1L–3.5L+

- Not inventing multimodal sensing — flex+EMG fusion is validated in research
- Innovation is packing it into a consumer-cost, home-usable wearable
- First affordable product to combine movement + muscle-activation sensing with real-time feedback

3 KEY FEATURES

- Dual-sensor fusion: kinematic (flex) + muscle-effort (EMG) — neither alone catches compensation patterns
- Real-time rep-quality feedback (not just a rep counter)
- On-device signal processing, BLE streaming to companion dashboard
- Session logging + progress dashboard for therapist review between visits

Small Device. Stronger Tomorrows.

SENSE • GUIDE • RECOVER



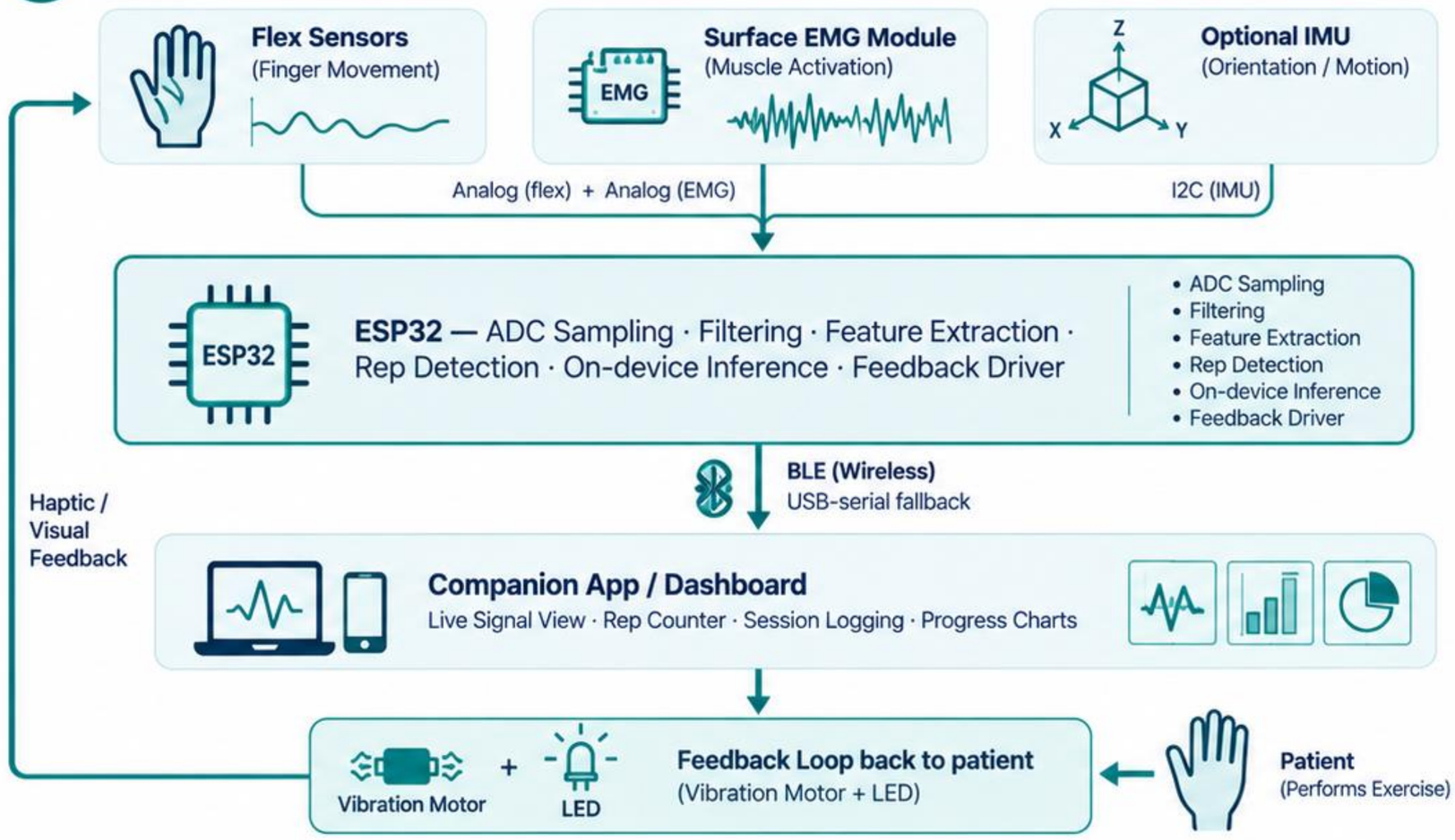
RehabGrip — Wearable Prototype

System Architecture & Technical Implementation

SENSE • PROCESS • FEEDBACK • RECOVER

From Signals to **Stronger Hands**

1 SYSTEM ARCHITECTURE



2 TECH STACK

Hardware / Embedded:
ESP32 (dual-core, WiFi/BLE), flex sensors (voltage-divider), MyoWare-class sEMG module, vibration motor + LED feedback actuators

Frontend:
Lightweight companion dashboard app — live sensor trace, rep counter, session summary, progress trend charts

Backend:
On-device signal processing (filtering, rectification, envelope extraction, feature extraction) running on ESP32; classification (Random Forest / SVM) on companion app for hackathon scope

Database:
Local session logging (compact JSON/binary per-repetition records — timestamp, ROM, EMG RMS, label)

APIs / Tools:
BLE GATT for streaming, USB-serial as wired demo fallback, scikit-learn (RF/SVM/LogReg) for classifier training

Demo Plan & Future Roadmap

FROM LIVE DEMO TO LASTING IMPACT

Build Today for **Stronger Tomorrows**

1 LIVE DEMO STRATEGY



Presenter wears the glove on stage and performs correct grip-and-release reps

Dashboard shows live flex + EMG signal trace and rep counter in real time

A deliberately incomplete rep is performed to trigger haptic/visual deviation feedback

Session summary + multi-session trend chart shown to close the demo

2 EXPECTED DELIVERABLES

- Working glove prototype (flex + EMG sensing, ESP32 acquisition)
- Real-time rep counting with correct/deviation classification
- Live companion dashboard (signal view, rep counter, session summary)
- Self-collected labeled dataset + trained classifier with evaluation metrics

3 FUTURE SCOPE & ROADMAP

SHORT TERM

- Add 2+ finger sensing, BLE streaming polish, per-user calibration
- Improve dashboard: live trace + session history

MID TERM

- Multiple exercise types beyond grip-and-release
- IMU integration to detect compensatory movement
- On-device ML inference — no companion app needed

LONG TERM

- Dedicated mobile app + cloud sync
- Therapist-facing analytics dashboard
- Clinical validation study with real patient population

0 – 6 Months → **6 – 18 Months** → **18+ Months**

24-Hour Grand Finale Build Scope & Feasibility

A FOCUSED MVP TODAY. A STRONGER TOMORROW.

Build Today
Enable **Brighter Tomorrows**

1 24-HOUR BUILD SCOPE (MVP LOCK)

MVP LOCK
Focused. Realistic. Buildable.

- One exercise: grip-and-release / finger flexion-extension
- One flex sensor (index finger) + one EMG channel
- Automatic repetition counting with correct/incomplete classification
- Real-time haptic or visual feedback
- Basic session dashboard — live trace, rep count, session summary
- Nothing beyond this attempted until the MVP is solid

2 COMPONENT CHECKLIST

HARDWARE

- [x] ESP32 DevKit board
- [x] Flex sensor, 2.2" (x2)
- [x] EMG sensor module (MyoWare-class) + electrode pads
- [x] Base fabric glove for mounting
- [x] Vibration motor + LED indicators
- [x] Li-ion/LiPo battery + charging module
- [x] Breadboard/perfboard, resistors, wires, connectors

Total MVP cost:
~₹5,700

EMBEDDED SYSTEM

- [x] ADC sampling of flex (voltage-divider) + EMG (analog envelope) channels
- [x] Filtering, rectification, envelope/RMS extraction (firmware-side)
- [x] Peak/valley rep segmentation + per-finger calibration routine
- [x] BLE streaming to dashboard, with USB-serial as wired demo fallback
- [x] Feedback actuator driver (vibration motor / LED) on detected deviation

3 SCALABILITY (post-hackathon, no new clinical research needed)

- Personalized per-user calibration baselines
- Additional exercises + additional fingers/EMG channels
- IMU integration for compensatory-movement detection
- On-device inference — fully standalone device, no companion app
- Dedicated mobile app replacing the dashboard prototype

From a 24-hour MVP to real-world impact.

Team Details

PEOPLE • PURPOSE • PROGRESS

Rehabilitation Today for a **Stronger Tomorrow**

Same Team. A Healthier Tomorrow.

TEAM LEADER

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SMALL STEPS STRONGER HANDS

THIRD-PARTY / OPEN-SOURCE DECLARATION

- **Arduino / ESP32 core framework** (open-source embedded firmware base)
- **scikit-learn** (Random Forest, SVM, Logistic Regression — classifier training)
- **BLE GATT protocol** (standard open wireless communication stack)
- **MyoWare-class sEMG sensor hardware/reference design** (commercial off-the-shelf module)